

Fig. 3: Structure of piezoelectric buzzer



Fig. 4: Piezoelectric buzzer

as computer speakers and portable radios. Piezoelectric speakers have several advantages over conventional loudspeakers: these are resistant to overloads which would normally destroy most high-frequency drivers, and they can be used without a crossover due to their electrical properties. There are also disadvantages: some amplifiers can oscillate when driving capacitive loads like most piezoelectrics, which results in distortion or damage to the amplifier. Additionally, their frequency response, in most cases, is inferior to that of the other technologies. This is why they are generally used in single-frequency (beeper) or non-critical applications.

Piezoelectric speakers can have extended high-frequency output, and this is useful in some specialised circumstances; for instance, sonar applications in which the piezoelectric variants are used as both the output devices (generating underwater sound) and the input devices (acting as the sensing components of underwater microphones). They have advantages in these ap-

plications, not the least of which is simple and solidstate construction which resists the effects of seawater better than, say, a ribbon-based device would.

Piezoelectric buzzers. Piezoelectric buzzers have a piezoelectric crystal mounted on a circular mounting plate

as shown in Fig. 3. Supporting struts are integrally formed in the mounting plate by punching strut-shaped portions in the plate and bending these portions such that they are substantially perpendicular to the plate. The struts are positioned on the nodal circle of the transducer so that the vibration of the transducer is not substantially impaired by the mounting arrangement.

An oscillator circuit is connected to provide a potential between the mounting plate and a first electrode positioned on the surface of the crystal opposite the mounting plate. A second electrode, positioned on the surface of the piezoelectric crystal adjacent to the first electrode, provides a feedback signal to the oscillator circuit. The unique buzzer design allows the buzzer components to be assembled by a one-step soldering process.

Horn speakers. Horn speakers are the oldest form of loudspeaker systems, having been used from very early on for cylinder recording players. As shown in Fig. 5, the part of the horn next to the speaker-cone driver is called the throat and the large part farthest away from the speaker cone is called the mouth. These use a shaped waveguide in front of or behind the driver to increase the directivity of the loudspeaker and to transform a small-diameter, high-pressure condition at the driver cone surface to a large-diameter, low-pressure condition at the mouth of the horn. This increases the sensitivity of the loudspeaker and focuses the sound over a narrower area. The size of the throat, mouth, the length of the horn, as well as the

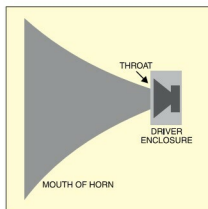


Fig. 5: Horn speaker

area expansion rate along it must be carefully chosen to match the drive to properly provide this transforming function over a range of frequencies (every horn performs poorly outside its acoustic limits, at both high and low frequencies).

The length and cross-sectional mouth area required to create a bass or sub-bass horn require a long horn.

Horn speakers can also be defined as loudspeakers that use a horn to increase the overall efficiency of the driving element, typically a diaphragm driven by an electromagnet. The horn itself is a passive component and does not amplify the sound from the driving element as such, but rather improves the coupling efficiency between the speaker driver and the air. The horn can be thought of as an acoustic transformer that provides impedance matching between the relatively dense diaphragm material (typically made of paper or, more recently, more exotic materials such as aluminium, titanium or PET film) and the air.

This is important because the difference in densities and motional characteristics of the air and the driving element is a mismatch.

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